Luis Felipe Casas Murillo

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SUMMARY

Broad-background researcher/engineer with knowledge in mechatronics, industrial automation, and cutting-edge AI/robotics. Highly motivated self-learner with a foundational, principles-based approach and continuous-improvement mindset; driven by discipline and a commitment to excellence in every task. Passionate about solving hard problems that benefit society. Fluent in Spanish and English, with basic proficiency in German.

EDUCATION

Philosophy Doctorate, Computer Science — Richardson, TX
University of Texas at Dallas | GPA: 4.0/4.0 — *Aug. 2023 – Current*

- Recipient of Louis Beecherl, Jr. Graduate Fellowship.
- Member of the Intelligent Robots and Vision Lab (IRVL) with research in dexterous manipulation, robotic control policies, robot grasping, generative AI, cross-embodiment learning, zero-shot transfer, computer vision and perception; utilizing PyTorch, JAX, Isaac Sim / Isaac Lab, MuJoCo, ROS/ROS2, robot design, precision sensors, electronics, Python, C++, reinforcement learning, imitation learning, kinematics, dynamics, motion planning, control theory, torque control, and task planning.
- Design, upgrade, and maintain robotic platforms for dexterous manipulation research; install and develop full compatibility for precision sensors (cameras, force-torque, IMU) for control policy development.
- RobotFingerPrint: Generative AI framework for unified gripper coordinate space, enabling multi-gripper grasp synthesis, and zero-shot generation to unknown robotic hands and objects (IROS 2025).
- MultiGripperGrasp: Open-source large-scale dataset with 30.4M grasps across 11 grippers using Isaac Sim for generalizable grasping, dexterous manipulation research, and benchmarking control policies (IROS 2024). 181 Stars Github repository.
- Benchmarked grasping and manipulation algorithms across diverse embodiments, developed pipelines to test performance of state-of-the-art robotic control policies, including simulation and real world scenarios using MuJoCo, Gazebo, Isaac Sim, and ROS/ROS2.
- Current projects: Cross-embodiment dexterous manipulation with zero-shot transfer; industry collaboration on novel robot hardware design for Frontier Mining.

Master of Science, Robotic Systems Engineering — Aachen, Germany
RWTH Aachen University | GPA: 3.9/4.0 — *Oct. 2019 – November 2022*

- Best of Class Awards (Multiple) and Member of the Dean's List.
- Digital Twin for Job Shops: Simulation and optimization of transportation agents in factories using reinforcement learning, Unity, digital twins, multi-agent systems, optimization, and C++/C# (LION 2023).
- Automated Renal Analysis: Computer vision model for hydronephrosis detection and precise kidney segmentation from ultrasound images, incorporating perception, TensorFlow/Keras, and Computer Vision (European Urology Open Science 2024).

Bachelor of Science, Mechatronics Engineering — Tegucigalpa, Honduras
UNITEC | GPA: 4.0/4.0 — *Jul. 2013 – Feb. 2018*

- Best of Class Awards (Multiple); Co-Founder, IEEE-RAS Student Chapter and president, Mechatronics Engineering Student Association (2017).
- Early Wildfire Detection System: Designed and deployed a wireless sensor network for real-time forest fire detection; developed custom sensor nodes, circuitry, enclosures, PCB layout, redundant communication protocol, and web application for live monitoring and alerts; emphasized embedded systems, wireless sensor networks, network management, and system testing/deployment (thesis).

PROFESSIONAL EXPERIENCE

University Lecturer & Lab Instructor — April 2023 – June 2023
UNITEC, Tegucigalpa (Honduras)

- Designed and delivered PLC and Industrial Networks courses for senior Mechatronics students, incorporating control theory, linear systems analysis, stability, controller design, task planning, and motor controls.
- Led hands-on labs using Siemens and Schneider hardware; guided full-cycle programming and commissioning from scratch for industrial applications in a cross-functional team environment, including automation, monitoring, and mechatronics integration.

AI Researcher Intern — November 2021 – March 2022
Hexafarms GmbH, Saarbrücken (Germany)

- Built 3D reconstruction pipeline using drone imagery and Structure-from-Motion for multimodal perception and object detection, incorporating physics/model representations.
- Developed deep learning models for yield prediction on edge devices, integrating sensors for real-time analysis and controls system implementation.

Mechatronics Engineer Aug. 2018 – May 2019

Ingelmec, Tegucigalpa (Honduras)

- Monitoring & Innovation Department + Fire Suppression Systems Team: Deployed datacenter monitoring systems with SCADA integration; performed on-site installation and network integration for clients.
- Maintained fire suppression systems; executed routine functional testing, troubleshooting, inspections, and system integration for reliability.

Laboratory Aide Jul. 2015 – Oct. 2017

UNILAB Multipurpose Laboratory, UNITEC (Honduras)

- Managed equipment and materials for chemistry, ecology labs: prepared chemical solutions, calibrated instruments for sensor accuracy and repeatability, and ensured safe lab conditions.
- Instructed 200+ students per trimester in laboratory safety protocols and experimental procedures.

TECHNICAL SKILLS

- **Robotic Manipulation Modeling and Control**: Dexterous Manipulation Policies, Robot Grasping, Cross-Embodiment Transfer, Modern Control Policies (Reinforcement Learning, Imitation Learning, Large Behavioral Models, Model Predictive Control), Modeling and Analysis (Forward/Inverse Kinematics, Torque Control, Spatial Vector Arithmetic, Stability, Controller Design, Motion Planning, Perception, Task Planning)
- **Machine Learning & AI**: Generative AI, 3D Reconstruction, Photogrammetry, Image Processing, Deep Learning, Controls System Development, Modeling, and Implementation
- **Frameworks & Tools**: PyTorch, JAX, TensorFlow/Keras, Isaac Sim / Isaac Lab, ROS/ROS2, MuJoCo, Gazebo, Unity, MATLAB/Simulink, Collaborative Coding (GitHub)
- **Robotics Hardware & Systems**: Robot Design, Precision Sensor Integration, Electronics, Embedded Systems, Wireless Sensor Networks, Circuit Design, PCB Layout, Motion Control Systems, CAD (i.e. Solid Works)
- **Software & Automation**: Python, C++/C#, PLC Programming, Industrial Networks, SCADA/Supervisory Systems, Human-Robot Interaction, Robotic Data Collection Pipelines, Edge Computing, System Integration, Algorithm Benchmarking

PUBLICATIONS

- **Materials Science Forum**, "Ale-HAND-Ra: Design and Development of a Low-Cost, Open-Source Humanoid Robotic Hand" Hillary Velásquez, Marcela Lagos, Fávell Núñez, L. F. Casas. 2026. doi:10.4028/p-wzT8b4
- **IROS**, "RobotFingerPrint: Unified Gripper Coordinate Space for Multi-Gripper Grasp Synthesis and Transfer," N. Khargonkar, L. F. Casas, B. Prabhakaran, Y. Xiang. 2025. arXiv:2409.14519.
- **IROS**, "MultiGripperGrasp: A Dataset for Robotic Grasping from Parallel Jaw Grippers to Dexterous Hands," L. F. Casas, N. Khargonkar, B. Prabhakaran, Y. Xiang. 2024. arXiv:2403.09841.
- **European Urology Open Science**, "Harnessing Artificial Intelligence for Enhanced Renal Analysis: Automated Detection of Hydronephrosis and Precise Kidney Segmentation," R. Alexa, et al. (incl. L. F. Casas). 2024. doi:10.1016/j.euros.2024.01.017.
- **Learning and Intelligent Optimization (LION).**, "Experimental Digital Twin for Job Shops with Transportation Agents," A. Gannouni, L. F. Casas, et al. 2023. doi:10.1007/978-3-031-44505-7_25.

Awards, Certifications and Professional Service

- Invited Speaker, International Engineering Week (May 2023) – UNITEC. Delivered talk titled "Artificial Intelligence Applied to Flexible Manufacturing Workshops".
- Judge, National Robotics Olympiad (WRO Honduras) (August 2018 – August 2019) – Second and Third editions, Honduras STEM Foundation.
- Deep Learning Specialization Certificate (March 2018) – deeplearning.ai (Coursera).
- Selected Participant, National Academic Week (February 2017) – Invitation-only event for the highest-achieving university students in Honduras, organized by Fundación Honduras Global at Zamorano Panamerican Agricultural School.
- Instructor, Welding and Splices Workshop (June 2016) – UNITEC. Received official recognition for delivering the workshop.